Small-prime split counts of the NIST/ANSI seed curves: null, sweep, held-out test

Study `research/nist_seed_split_null_20261029`, 29 October 2026 UTC. Manuscript draft: `papers/nist-seed-split-null/`. Procedural deviations D1–D3 are in `DEVIATIONS.md`.

Question. Do the published prime-field curves have fewer split primes (odd $\ell \leq 691$ with $t^2 - 4p$ a nonzero square mod ℓ) than unpublished prime-order seed candidates?

Prime-order confound (derived). All 19 named curves have prime order. The exact Chebotarev null conditioned on prime order has mean 59.7–60.2 and s.d. 5.5. Prime order moves the mean by only 0.5–1.4 splits, and small factors of the twist order force 0–4 splits per curve. Validation at 48 bits printed aggregates only and kept no rows (D2).

Cutoff, exploratory, on the named 19. The 25/25 within-field score holds only for $B \in \{691, 701, 709, 769\}$. Choosing B over 655 prime cutoffs gives $p = 0.0055$ (Monte Carlo, n = 20,000) or 0.0065 (exact permutation, 4,480 allocations). That correction does not cover the earlier choice of statistic.

Held-out test, preregistered in Markdown before data (`bd5d146b`). Over 42 unpublished draft-eligible curves outside the named eleven, held-out curves exceed published ones in 0.574 of 102 within-field pairs, one-sided permutation $p = 0.29$. The association does not replicate. The held-out percentiles fit the null: KS $D = 0.153$ against a 5% critical value of 0.210, n = 42. One unpublished 192-bit curve has 44 splits. The one surviving extreme is prime239v1 (42 splits, 0.06th percentile), with a null tail of about 0.05 among the 61 eligible curves examined.

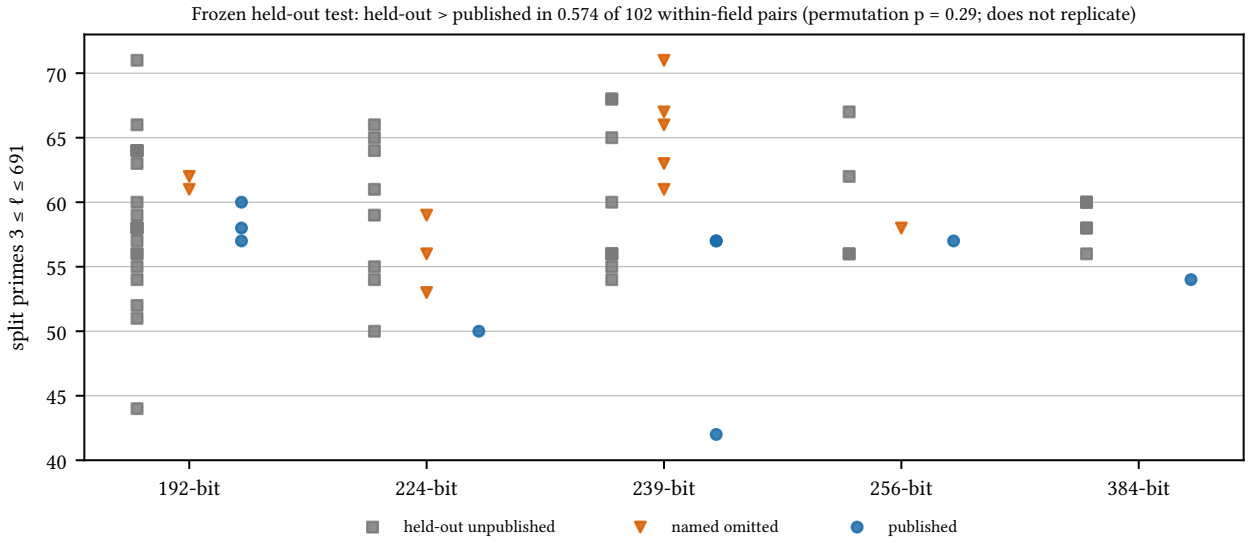

Figure 1: Split counts at $B = 691$ by field. Source: `runs/heldout.json`, `runs/rows.jsonl`. Drawn by `figures.py`.

Scope. No hypothesis status changes, and no ECDLP mechanism is known. The selection-rule problem stays open.